Quiz Application Using File Handling in C++

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```
1 0 1 0 1 1 1 1 1 1
0 1 1 1 1 0 0 0 1 0
1 1 0 0 0 0 1 0 1 1
1 0 0 1 0 1 0 1 1 1
0 0 1 0 1 0 1 1 1 0
0 0 0 1 0 1 0 0 0 0
1 0 1 0 1 1 1 0 1 1
0 0 1 1 1 0 0 1 0 0
U S I N E S S 1 1 1
1 1 1 1 1 1 0 0 0 1
0 1 0 0 0 0 1 0 1 0
```

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1 0 1 0 1 1 1 1 1 1
0 1 1 1 1 0 0 0 1 0
1 1 0 0 0 0 1 0 1 1
1 0 0 1 0 1 0 1 1 1
0 0 1 0 1 0 1 1 1 0
0 0 0 1 0 1 0 0 0 0
1 0 1 0 1 1 1 0 1 1
0 N E T W O R K 0 0
1 1 0 0 0 0 1 1 1 1
1 1 1 1 1 1 0 0 0 1
0 1 0 0 0 0 1 0 1 0
```

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1 0 1 0 1 1 1 1 1 1
0 1 1 1 1 0 0 0 1 0
1 1 0 0 0 0 1 0 1 1
1 0 0 1 0 1 0 1 1 1
0 0 1 0 1 0 1 1 1 0
0 0 0 1 0 1 0 0 0 0
1 0 1 0 1 1 1 0 1 1
I N T E R N E T 0 0
1 1 0 0 0 0 1 1 1 1
1 1 1 1 1 1 0 0 0 1
0 1 0 0 0 0 1 0 1 0
```

```
1 0 1 0 1 1 1 1 1 1
0 1 1 1 1 0 0 0 1 0
1 1 0 0 0 0 1 0 1 1
1 0 0 1 0 1 0 1 1 1
0 0 1 0 1 0 1 1 1 0
0 0 0 1 0 1 0 0 0 0
1 0 1 0 1 1 1 0 1 1
0 0 1 E R R O R 0 0
1 1 0 0 0 0 1 1 1 1
1 1 1 1 1 1 0 0 0 1
0 1 0 0 0 0 1 0 1 0
```

Introduction

Problem Statement: In educational settings, quizzes are a common method for assessing knowledge. However, managing quizzes, tracking user scores, and allowing users to register and log in can be cumbersome without a systematic approach.

OOP Concepts

Classes and Objects

Encapsulation

Inheritance

Static Members

Method Overloading

Data Abstraction

C++ Libraries Used

iostream

cstring

string

fstream

sstream

vector

algorithm

```
0 1 1 1 1 1 1 0 1 0 1
1 1 0 0 0 1 0 1 1 1 1
0 0 0 1 0 1 1 1 0 0 0
1 0 1 0 1 1 1 0 0 1 0
0 1 0 1 1 1 0 0 1 0 1
1 0 1 0 0 0 0 0 0 1 0
0 1 1 1 0 1 1 0 1 0 1
1 1 0 0 1 0 0 D A T A
0 0 0 1 1 1 1 1 0 L E
1 1 1 0 0 0 1 1 1 1 1
0 0 0 1 0 1 0 1 0 0 0

0 1 1 1 1 1 1 0 1 0 1
1 1 0 0 0 1 0 1 1 1 1
0 0 0 1 0 1 1 1 0 0 0
1 0 1 0 1 1 1 0 0 1 0
0 1 0 1 1 1 0 0 1 0 1
1 0 1 0 0 0 0 0 0 1 0
0 1 1 1 0 1 1 0 1 0 1
1 1 0 0 1 0 0 0 M O N
0 0 0 1 1 1 1 1 0 0 0
1 1 1 0 0 0 1 1 1 1 1
0 0 0 1 0 1 0 1 0 0 0

0 1 1 1 1 1 1 0 1 0 1
1 1 0 0 0 1 0 1 1 1 1
0 0 0 1 0 1 1 1 0 0 0
1 0 1 0 1 1 1 0 0 1 0
0 1 0 1 1 1 0 0 1 0 1
1 0 1 0 0 0 0 0 0 1 0
0 1 1 1 0 1 1 0 1 0 1
1 1 0 0 1 0 0 H A C K
0 0 0 1 1 1 1 1 A T T
1 1 1 0 0 0 1 1 1 1 1
0 0 0 1 0 1 0 1 0 0 0

0 1 1 1 1 1 1 0 1 0 1
1 1 0 0 0 1 0 1 1 1 1
0 0 0 1 0 1 1 1 0 0 0
1 0 1 0 1 1 1 0 0 1 0
0 1 0 1 1 1 0 0 1 0 1
1 0 1 0 0 0 0 0 0 1 0
0 1 1 1 0 1 1 0 1 0 1
1 1 0 0 1 0 0 0 1 D A
0 0 0 1 1 1 1 1 0 0 0
1 1 1 0 0 0 1 1 1 1 1
0 0 0 1 0 1 0 1 0 0 0
```

Tasks Performed

Students: Registration, Login, Taking a quiz, View scores, View student details, View leader board

Admin: Login, Add questions, View scores, View leader board, Remove questions

```
C:\Windows\System32\cmd.e

Microsoft Windows [Version 10.0.22631.4317]
(c) Microsoft Corporation. All rights reserved.

D:\CODES-C++>a.exe

LOGIN AS:
1. Student
2. Admin
3. Exit
Your Choice: 1

1. Register
2. Login
Enter Your Choice: 2
ENTER USERNAME: aryan
ENTER PASSWORD: abc
LOGGED IN

1. TAKE QUIZ
2. VIEW SCORE
3. VIEW STUDENT DETAILS
4. VIEW LEADERBOARD
5. Exit
YOUR CHOICE: 1

What is the process by which water vapor turns into liquid water?
a) Evaporation
b) Condensation
c) Sublimation
d) Precipitation

YOUR ANSWER: b
What is the capital of France?
a) Berlin
b) Madrid
c) Paris
d) Rome

YOUR ANSWER: c
Who was the first President of the United States?
a) Abraham Lincoln
b) Thomas Jefferson
c) George Washington
d) John Adams

YOUR ANSWER: a
Who wrote the play 'Romeo and Juliet'?
a) Charles Dickens
b) William Shakespeare
```

```
C:\Windows\System32\cmd.e

b) William Shakespeare
c) Mark Twain
d) Leo Tolstoy

YOUR ANSWER: b
Which is the largest planet in our solar system?
a) Earth
b) Mars
c) Jupiter
d) Saturn

YOUR ANSWER: a
Which gas do plants absorb from the atmosphere during photosynthesis?
a) Oxygen
b) Carbon Dioxide
c) Nitrogen
d) Hydrogen

YOUR ANSWER: b
What is the currency of Japan?
a) Yen
b) Won
c) Dollar
d) Peso

YOUR ANSWER: a
Which element has the chemical symbol 'O'?
a) Gold
b) Oxygen
c) Osmium
d) Silver

YOUR ANSWER: a
What is the smallest continent by land area?
a) Europe
b) Australia
c) Antarctica
d) South America

YOUR ANSWER: b
Which ocean is the largest in the world?
a) Atlantic Ocean
b) Indian Ocean
c) Arctic Ocean
d) Pacific Ocean

YOUR ANSWER: d
YOUR SCORE IS: 7
```

```
0 1 1 1 1 1 1 0 1 0 1
1 1 0 0 0 1 0 1 1 1 1
0 0 0 1 0 1 1 1 0 0 0
1 0 1 0 1 1 1 0 0 1 0
0 1 0 1 1 1 0 0 1 0 1
1 0 1 0 0 0 0 0 0 1 0
0 1 1 1 0 1 1 0 1 0 1
1 1 0 0 1 0 0 D A T A
0 0 0 1 1 1 1 1 0 L E
1 1 1 0 0 0 1 1 1 1 1
0 0 0 1 0 1 0 1 0 0 0

0 1 1 1 1 1 1 0 1 0 1
1 1 0 0 0 1 0 1 1 1 1
0 0 0 1 0 1 1 1 0 0 0
1 0 1 0 1 1 1 0 0 1 0
0 1 0 1 1 1 0 0 1 0 1
1 0 1 0 0 0 0 0 0 1 0
0 1 1 1 0 1 1 0 1 0 1
1 1 0 0 1 0 0 0 M O N
0 0 0 1 1 1 1 1 0 0 0
1 1 1 0 0 0 1 1 1 1 1
0 0 0 1 0 1 0 1 0 0 0

0 1 1 1 1 1 1 0 1 0 1
1 1 0 0 0 1 0 1 1 1 1
0 0 0 1 0 1 1 1 0 0 0
1 0 1 0 1 1 1 0 0 1 0
0 1 0 1 1 1 0 0 1 0 1
1 0 1 0 0 0 0 0 0 1 0
0 1 1 1 0 1 1 0 1 0 1
1 1 0 0 1 0 0 H A C K
0 0 0 1 1 1 1 1 A T T
1 1 1 0 0 0 1 1 1 1 1
0 0 0 1 0 1 0 1 0 0 0

0 1 1 1 1 1 1 0 1 0 1
1 1 0 0 0 1 0 1 1 1 1
0 0 0 1 0 1 1 1 0 0 0
1 0 1 0 1 1 1 0 0 1 0
0 1 0 1 1 1 0 0 1 0 1
1 0 1 0 0 0 0 0 0 1 0
0 1 1 1 0 1 1 0 1 0 1
1 1 0 0 1 0 0 0 1 D A
0 0 0 1 1 1 1 1 0 0 0
1 1 1 0 0 0 1 1 1 1 1
0 0 0 1 0 1 0 1 0 0 0
```

C:\Windows\System32\cmd.e

```
YOUR SCORE IS: 7

1. TAKE QUIZ
2. VIEW SCORE
3. VIEW STUDENT DETAILS
4. VIEW LEADERBOARD
5. Exit
YOUR CHOICE: 3


USERNAME: aryan
NAME: aryan sharma

1. TAKE QUIZ
2. VIEW SCORE
3. VIEW STUDENT DETAILS
4. VIEW LEADERBOARD
5. Exit
YOUR CHOICE: 4

NAME: aditya thakur
USERNAME: aditya
SCORE: 7

NAME: aryan sharma
USERNAME: aryan
SCORE: 7

NAME: dhruav mathur
USERNAME: dhruav
SCORE: 6

NAME: atulit singh
USERNAME: atulit
SCORE: 5

NAME: ashish ohol
USERNAME: ashish
SCORE: 4


1. TAKE QUIZ
2. VIEW SCORE
3. VIEW STUDENT DETAILS
4. VIEW LEADERBOARD
5. Exit
YOUR CHOICE: 5
```

C:\Windows\System32\cmd.e

```
LOGIN AS:
1. Student
2. Admin
3. Exit
Your Choice: 2
ENTER USERNAME: dhruav
ENTER PASSWORD: abc123
LOGGED IN

1. ADD QUESTIONS
2. VIEW SCORES
3. VIEW LEADERBOARD
4: REMOVE QUESTIONS
5. Exit
YOUR CHOICE: 2

dhruav          dhruav mathur           6
ashish          ashish ohol             4
atulit          atulit singh            5
aditya          aditya thakur           7
anuj            anuj singh              3
aryan           aryan sharma            7

1. ADD QUESTIONS
2. VIEW SCORES
3. VIEW LEADERBOARD
4: REMOVE QUESTIONS
5. Exit
YOUR CHOICE: 3

NAME: aditya thakur
USERNAME: aditya
SCORE: 7

NAME: aryan sharma
USERNAME: aryan
SCORE: 7

NAME: dhruav mathur
USERNAME: dhruav
SCORE: 6

NAME: atulit singh
USERNAME: atulit
SCORE: 5

NAME: ashish ohol
USERNAME: ashish
```

```
C:\Windows\System32\cmd.e

USERNAME: dhruav
SCORE: 6

NAME: atulit singh
USERNAME: atulit
SCORE: 5

NAME: ashish ohol
USERNAME: ashish
SCORE: 4


1. ADD QUESTIONS
2. VIEW SCORES
3. VIEW LEADERBOARD
4: REMOVE QUESTIONS
5. Exit
YOUR CHOICE: 4

1. Which is the largest planet in our solar system?
2. What is the capital of France?
3. Who wrote the play 'Romeo and Juliet'?
4. Which element has the chemical symbol 'O'?
5. What is the smallest continent by land area?
6. Which gas do plants absorb from the atmosphere during photosynthesis?
7. Which ocean is the largest in the world?
8. What is the currency of Japan?
9. What is the process by which water vapor turns into liquid water?
10. Who was the first President of the United States?
ENTER QUESTION NUMBER TO DELETE: 8

1. ADD QUESTIONS
2. VIEW SCORES
3. VIEW LEADERBOARD
4: REMOVE QUESTIONS
5. Exit
YOUR CHOICE: 5


LOGIN AS:
1. Student
2. Admin
3. Exit
Your Choice: 3


Thank you
Exiting...
D:\CODES-C++>
```

```
0 1 1 1 1 1 1 0 1 0 1
1 1 0 0 0 1 0 1 1 1 1
0 0 0 1 0 1 1 1 0 0 0
1 0 1 0 1 1 1 0 0 1 0
0 1 0 1 1 1 0 0 1 0 1
1 0 1 0 0 0 0 0 0 1 0
0 1 1 1 0 1 1 0 1 0 1
1 1 0 0 1 0 0 D A T A
0 0 0 1 1 1 1 1 0 L E
1 1 1 0 0 0 1 1 1 1 1
0 0 0 1 0 1 0 1 0 0 0

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1 1 0 0 1 0 0 0 M O N
0 0 0 1 1 1 1 1 0 0 0
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0 1 1 1 0 1 1 0 1 0 1
1 1 0 0 1 0 0 H A C K
0 0 0 1 1 1 1 1 A T T
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1 0 1 0 0 0 0 0 0 1 0
0 1 1 1 0 1 1 0 1 0 1
1 1 0 0 1 0 0 0 1 D A
0 0 0 1 1 1 1 1 0 0 0
1 1 1 0 0 0 1 1 1 1 1
0 0 0 1 0 1 0 1 0 0 0
```

References

GEEKSFORGEEKS : https://www.geeksforgeeks.org/object-oriented-programming-in-cpp/

W3SCHOOLS: https://w3schools.com/cpp/cpp_oop.asp

PPTS OF OBJECT ORIENTED IN C++

Stack Overflow

Conclusion

By utilizing file handling for data persistence, the application provides a seamless experience for both users and administrators.
This project not only addresses the problem of managing quizzes but also serves as a practical example of applying OOP concepts in C++.

```
0 1 1 1 1 1 1 0 1 0 1
1 1 0 0 0 1 0 1 1 1 1
0 0 0 1 0 1 1 1 0 0 0
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1 1 0 0 1 0 0 D A T A
0 0 0 1 1 1 1 1 0 L E
1 1 1 0 0 0 1 1 1 1 1
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1 1 0 0 1 0 0 0 M O N
0 0 0 1 1 1 1 1 0 0 0
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1 1 0 0 1 0 0 H A C K
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1 0 1 0 0 0 0 0 0 1 0
0 1 1 1 0 1 1 0 1 0 1
1 1 0 0 1 0 0 0 1 D A
0 0 0 1 1 1 1 1 0 0 0
1 1 1 0 0 0 1 1 1 1 1
0 0 0 1 0 1 0 1 0 0 0
```